

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 2 Lecture - Introduction to Differentiation

LAST WEEK...

- For a function, $y = f(x)$, that describes some process we are interested in, in one dimension, the derivative:

$$f'(x) = \frac{dy}{dx}$$

- It is interpreted as the rate of change of 'y' with respect to 'x'.
- Lets imagine we are observing this change over a discrete interval, such that we could write this as an approximation:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

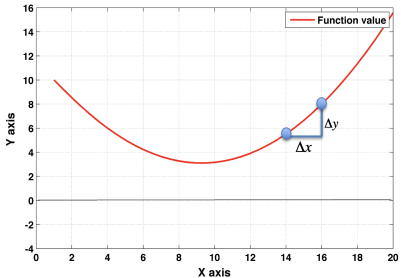
Last Week...

- Last week we looked at the quadratic function:

$$y = \sqrt{x} + \left(\frac{x-10}{3}\right)^2$$

- We want the derivative at $x = 16$.
- Assume, for now, no analytical function can be derived.
- **Euler's Backward Method**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_x - y_{x-\Delta x}}{\Delta x}$$



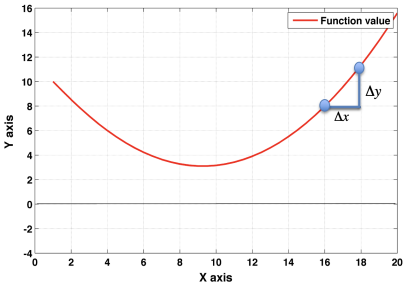
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- **Euler's Forward Method**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_x}{\Delta x}$$



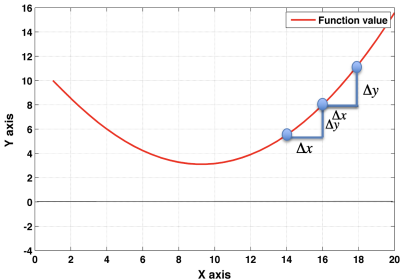
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- Assume, for now, no analytical function can be derived.
- **Central Differences Method**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$



Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the concept of differentiation and explain its significance in calculus.

(2) Identify and interpret common notation used in differentiation.

(3) Apply common laws of differentiation to solve theoretical and real-world problems.

Example: Radioactive Decay

- The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay rate.

- It basically says that the change in the number of nuclei in time is proportional to the number or nuclei present.
- This is a differential equation and it has a solution:

$$\begin{aligned} N(t) &= N_0 \exp(-\lambda t) \\ &= N_0 e^{-\lambda t} \end{aligned}$$

where N_0 is the number of nuclei in the sample. Note that I sometimes write e^x as $\exp(x)$, they mean the same thing.

Example: Radioactive Decay

Assumed time step, or time increment, to calculate the numerical derivative

	Time step	Numerical Derivative	Analytical Derivative	% error
	4000000			
Number steps from t=0	4.00E+21			
1	3.9632E+21	-9.19939E+12	-9.15694E+12	0.46138634
2	3.92674E+21	-9.11476E+12	-9.0727E+12	0.46138634

Number of atoms decreases according to our exponential function

$$N = N_o \exp(-\lambda t) = N_o e^{-\lambda t}$$

Given analytical derivative

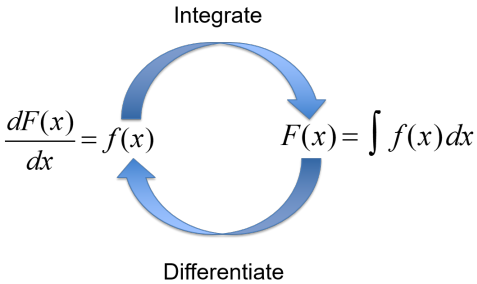
$$\frac{dN}{dt} = -\lambda N_o \exp(-\lambda t); \frac{dN}{dt} = -\lambda N$$

Approximate with numerical derivative

$$\frac{dN}{dt} = \lim_{\Delta x \to 0} \frac{\Delta N}{\Delta t} \equiv \frac{N_{current} - N_{previous}}{t_{current} - t_{previous}}$$

What is calculus?

- Its is also a language and a way of reasoning.
- In this course, we will learn to use specific notation when we construct that reasoning and solve several physical laws.
- Differentiation goes from right to left!



Differentiation

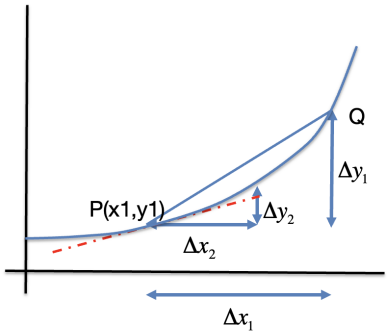
- Whilst we can find the slope of the curve 'graphically', or 'numerically', we want the most accurate solution if possible.

$$\frac{\Delta y_1}{\Delta x_1} \neq \frac{\Delta y_2}{\Delta x_2}$$

- We need a formulaic way of deriving this value.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

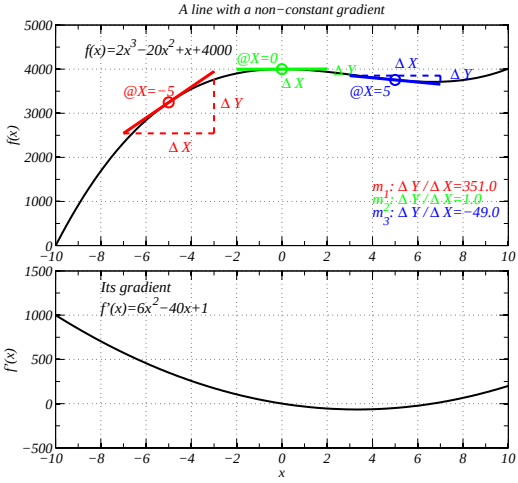
- There are rules we can rely on!



Differentiation

- If we could derive expressions for the approximation, what if we could find a general rule?

$$y' = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$



Example: Linear function

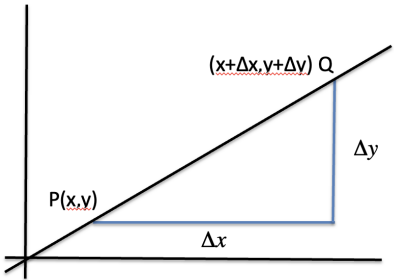
- Let's derive a derivative for a straight line.
- The equation of a straight line can be expressed as the following:

$$y = ax$$

where a is a constant.

- Lets find an expression for $\frac{\Delta y}{\Delta x}$.
- If on the graph we start at (x, y) and shift Δx on the x-axis

$$\begin{aligned} y + \Delta y &= a(x + \Delta x) \\ &= ax + a\Delta x \end{aligned}$$



Example: Linear function

- Let's take away $y = ax$

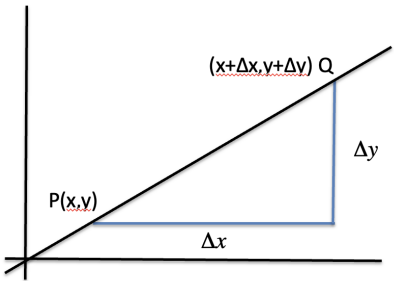
$$y + \Delta y - y = ax + a\Delta x - ax$$
$$\Delta y = a\Delta x$$

- Dividing by Δx gives us

$$\frac{\Delta y}{\Delta x} = a$$

- If $\Delta x \to 0$ then $\frac{\Delta y}{\Delta x} \to \frac{dy}{dx}$ and then the analytical derivative is

$$\frac{dy}{dx} = a$$



Example: Quadratic function

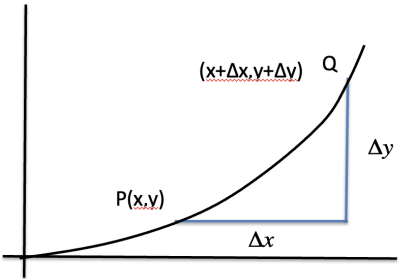
- Let's derive a derivative for a quadratic function.
- The equation of a quadratic function can be expressed as the following:

$$y = ax^2$$

where a is a constant.

- Lets find an expression for $\frac{\Delta y}{\Delta x}$.
- If on the graph we start at (x, y) and shift Δx on the x-axis

$$\begin{aligned} y + \Delta y &= a(x + \Delta x)^2 \\ &= ax^2 + 2ax(\Delta x) + a(\Delta x)^2 \end{aligned}$$



Example: Quadratic function

- Let's take away $y = ax^2$

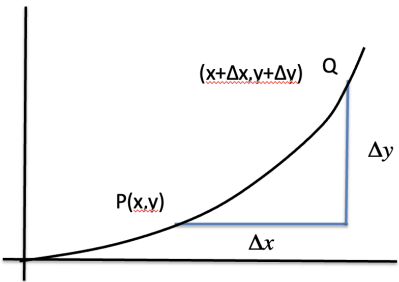
$$y + \Delta y - y = ax^2 + 2ax(\Delta x) + a(\Delta x)^2 - ax^2$$
$$\Delta y = 2ax(\Delta x) + a(\Delta x)^2$$

- Dividing by Δx gives us

$$\frac{\Delta y}{\Delta x} = 2ax + a\Delta x$$

- If $\Delta x \to 0$ then $\frac{\Delta y}{\Delta x} \to \frac{dy}{dx}$ and then the analytical derivative is

$$\frac{dy}{dx} = 2ax$$



Differentiation - Common Laws

- Based on that initial analysis, a rule which we can apply to any such function falls out in that if $y = x^n$ then

$$\frac{dy}{dx} = nx^{n-1}$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
a	0
ax	a
ax^2	$2ax$
ax^3	$3ax^2$
ax^4	$4ax^3$
x^n	nx^{n-1}
ax^n	anx^{n-1}

Differentiation - Common Laws

- We can apply similar procedures to other common functions and start populating a list of rules.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!
- In class this week you will practice using all of these, and also have questions on the weekly assessment related to them.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Differentiation - Common Laws

- What happens when we have a sum of functions?

$$y = 5x + e^x$$

- Well luckily, the derivative of the sum is the sum of the derivatives!
- This means we can just differentiate each term individually

$$\frac{dy}{dx} = 5 + e^x$$

- We'll see next week what happens when we have more complicated combinations of functions.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- The process by which a nucleus of an unstable atom emits energy to reach a more stable state. The abundance of atoms in these different states are described using the following function:

$$N(t) = N_0 \exp(-\lambda t)$$

  where N_0 is the number of nuclei in the sample.

- For which we can derive two important derivative functions:

$$\frac{dN}{dt} = -\lambda N_0 \exp(-\lambda t) \qquad \frac{dN}{dt} = -\lambda N$$

- Notice how we have used the rule for obtaining derivatives of exponential functions and then simplified to show the instantaneous decay is proportional to the existing abundance.
- Why highlight 'instantaneous'?

Your Growing Skills Library

We might have a function

$$y = f(x)$$

We want the derivative

$$\frac{dy}{dx}$$

The function we derive represents change over an infinitesimally small region

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

